COMP396 Weekly Meeting Minutes

(Week 2)

Date& Time: Friday, October 3rd, 2025, 14:00-17:00

Location: Online (Tencent meeting)

Note Taker: Zhang Olivia

Attendees: Zhang Olivia, Xue Yudan, Zhao Saria, Wong Haw

1. Agenda:

- Check the completion status of the first week's basic tasks (A1–A5).
- Share analysis findings of the sample strategies (rsi_contrarian, bbands_contrarian) (A3).
- Discuss specific implementation methods of Limit Orders based on research (A4).
- Conduct initial brainstorming for the three independent strategies.
- Plan "**Phase 2: Core Feature Implementation**" and assign new tasks.

2. Discussion & Decisions:

2.1. Task Review and Result Confirmation

The team confirmed that all the basic tasks (A1–A5) assigned in last week's meeting were completed on time.

- **Environment Setup (A1):** All members successfully set up the local VS Code development

environment and finished Git and private GitHub repository settings.

- **Core Concept Study (A2 & A4):** All members learned about budget limits, bankruptcy rules, and the mechanisms of market and limit orders.

- **Sample Strategy Analysis (A3):**

  - Xue Yudan and Zhang Olivia presented an analysis of the rsi_contrarian strategy, pointing out its core logic is trading against the RSI indicator, buying when oversold and selling when overbought.

  - Wong Haw and Zhao Saria presented an analysis of the bbands_contrarian strategy, summarizing its use of Bollinger Bands' upper and lower limits as reversal trading signals.

  - In conclusion, these sample strategies provided us with a valuable framework, especially for signal generation and position management.

- **Collaborative Report Structure (A5):** Zhang Olivia has created a shared report document in OneDrive and jointly set up the outline of the main parts required for the final report with Zhao Saria (see Appendix).

2.2. Core Knowledge Deepening and Study Plan

Since the lecturer could not give an in-person class this week and only uploaded teaching materials on Canvas, the team made the following self-study plan:

- **Focused PPT Self-study:** The PPT Lecture 03: The Backtester rules & Order Types was uploaded to Canvas. All members agreed that we must study this PPT carefully, as it is directly related to tasks A2 and A4. Special attention should be paid to key points such as slippage, limit order execution logic, and other details.

- **Backtrader Framework Review:** Although we already gained some understanding of Backtrader through exercises, the team agreed that we need a more systematic review to prepare for more complex strategy development later. Therefore, we decided to fully review last semester's COMP226 official materials on Backtrader to strengthen and expand our knowledge base.

2.3. Next Phase Plan: From Basics to Practice

After completing the basic preparation, the team's focus will shift fully to the **"Phase 2"** planned in the **"Future Task List"**.

- **Core Feature Implementation (Phase 2):** The team agreed that it is time to move from theory to practice. Next week, we will focus on completing all the exercises required for Phase 2, including expanding Data Processing capabilities (OHLCV), integrating technical indicators and learning parameter optimization.

3. Task Allocation:

To ensure the efficient advancement of the project and prepare well for the first assignment (due Nov 13), the team made the following task allocation:

ID	Module	Task Description	Owner	Due Date
B1	Core Knowledge Deepening	Self-study Lecture 3 on Canvas, ensuring full understanding of budget limits, bankruptcy, slippage, and two types of order rules.	All Members	Oct.08
B2	Core Feature Implementation	Subtask B2.1: Finish **Exercises #5** and **#6**, extend data processing ability from closing price to full OHLCV format.	All Members	Oct.08
		Subtask B2.2: Finish **Exercise #10**, study and practice how to use pandas-ta library for technical indicators.	All Members	Oct.08
		Subtask B2.3: Finish **Exercise #9**, learn and apply Backtrader's optstrategy for basic parameter optimization.	All Members	Oct.08
B3	Framework Document Review	Complete review of Backtrader official documents, focusing on strategies, data sources, orders, indicators, and analyzers.	All Members	Oct.10
B4	Initial Strategy Ideas	Based on previous learning, each member prepares an initial individual strategy idea(e.g., core logic, commonly used indicators) for discussion in the next meeting.	All Members	Oct.10

4. Current Challenges and Issues

- **Knowledge Integration Challenge:** The main challenge now is how to integrate

scattered knowledge from documents, course PPTs, and coding exercises into a clear, effective, and logically strict trading strategy.

- **Time Management and Project Pressure**
  - The deadline for the first assignment is less than five weeks away.
  - We must handle both technical practice and strategy creation at the same time.
  - On one hand, we need to complete the core exercises and systematically review Backtrader to fill knowledge gaps. On the other hand, the report requires us to submit a complete strategy design with initial testing and out-of-sample results.
  - We worry that too much time may be consumed by coding details and technical difficulties, leaving less time for strategy design, data analysis, parameter optimization, and result evaluation.

5. Supervisor Question

- Everything is going smoothly and according to the planned timeline. No immediate obstacles or issues have been found.

6. Next Meeting Arrangement

- **Date & Time:** Friday, October 10th, 2025, 14:00-17:00
- **Location:** Sydney Jones Library / Abercromby Study Room 1
- **Key Agenda Items:**

- Check the completion of “Phase 2 - core feature tasks” (B2).
- Share core gains from Backtrader document review (B3).
- Each member presents and discusses their initial individual strategy ideas (B4).
- Plan and assign “Phase 3: Risk and Portfolio Management”.

Appendix:

COMP396 设计报告 - 结构大纲

Part 1: Individual Strategy Designs (独立策略设计)

(总页数限制: 15 页)

1.1 Strategy One: [策略一名称, e.g., Bollinger Bands Mean-Reversion Strategy]

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- **1.1.1 Strategy Overview & Rationale (策略概述与基本原理)**
  - **Strategy Type:** 明确策略类型 (e.g., 均值回归、趋势跟踪、套利等)。
  - **Market Conditions:** 描述该策略在何种市场条件下最可能盈利 (e.g., 高波动性横盘市场、稳定上升趋势市场)。
  - **Time-frame:** 阐述策略的交易频率和持仓周期 (e.g., 日内交易、周度调仓)。
- **1.1.2 Data Analysis & Market Condition Identification (数据分析与市场条件验证)**
  - **Applicable Series:** 识别并论证哪些数据序列适合本策略。
  - **Evidence of Viability:** 提供数据证据来支持策略的有效性。例如，如果是均值回归策略，需提供所选序列存在均值回归特性的统计证据 (e.g., ADF 检验、半衰期计算)。
  - **Indicator Analysis:** 展示与策略相关的技术指标在历史数据上的表现 (e.g., RSI 指标在超买/超卖区的分布情况)。
- **1.1.3 Technical Specification (技术规格说明)**
  - **Entry Conditions:** 用清晰的伪代码或 Python 代码片段，详细描述具体的入场条件 (e.g., IF close < lower_bollinger_band AND RSI < 30 THEN BUY)。

    - **Exit Conditions:** 同样，用清晰的伪代码或代码描述出场条件（e.g., IF close > middle_bollinger_band OR holding_days > 5 THEN SELL)。

- **1.1.4 Preliminary Testing & Out-of-Sample Results (初步测试与样本外结果)**
    - **Data Partitioning:** 说明如何将第一部分数据划分为训练集和测试集。
    - **Parameter Optimization:**
        - **Parameters & Ranges:** 列出策略中的所有可调参数及其优化范围（e.g., moving_average_period: [10, 20, 30])。
        - **Optimization Method:** 描述选择参数组合的方法（e.g., 网格搜索）和评估标准（e.g., 选择夏普比率最高的参数组合)。
    - **Out-of-Sample Performance:** 展示策略在**测试集**上的回测结果，强调这是未经优化的样本外表现，并附上关键性能指标（e.g., PD Ratio, Max Drawdown)。
- **1.1.5 Future Development Plan (未来开发计划)**
    - 描述在第二学期将如何继续改进此策略，以及如何利用第二部分数据进行验证。

1.2 Strategy Two: [策略二名称, e.g., Dual Moving Average Trend-Following Strategy]

(重复以上 1.1.1 至 1.1.5 的结构)

1.3 Strategy Three: [策略三名称, e.g., Volatility Breakout Strategy]

(重复以上 1.1.1 至 1.1.5 的结构)

Part 2: The Combined Strategy (组合策略)

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- **2.1 Portfolio Construction & Capital Allocation (投资组合构建与资本分配)**
    - **Strategy Combination Logic:** 解释如何将三个独立策略融合成一个整体策略。说明策略间是否存在交易重叠，以及如何处理。
    - **Capital Allocation Model:** 详细说明 100 万英镑总资本如何在三个策略间进行分配（e.g., 等权重分配、基于风险的动态分配等)。
- **2.2 Global Risk Management Framework (全局风险管理框架)**
    - **Trade-Level Controls:** 描述单笔交易层面的风控措施（e.g., 是否使用止损单/止盈单)。
    - **Strategy-Level Controls:** 描述策略层面的风控，例如当某个策略持续亏损或最大回撤超过阈值时，是否会动态调整其权重或暂停该策略。

    - **Portfolio-Level Bankruptcy Mitigation:** 阐述如何从投资组合层面管理和规避破产风险 。
- **2.3 Position Sizing Methodology (头寸规模确定方法)**
    - **Cross-Strategy Sizing:** 解释在不同策略之间如何确定相对头寸大小。
    - **Individual Strategy Sizing:** 如果某个策略有其独特的头寸管理方法 (e.g., 固定分数法、波动率均衡法)，在此处进行说明。
- **2.4 Future Development Plan for the Combined Model (组合模型的未来开发计划)**
    - 阐述在第二学期将如何继续优化组合策略的资本分配和风险管理模块。

Part 3: Breakdown of Teamwork (团队成员贡献说明)

(每位成员 1 页)

- **3.1 Xue Yudan's Contributions**
    - *(在此处详细描述在背景研究、编码实现、策略构思、报告撰写等方面的具体贡献)*
- **3.2 Zhao Saria's Contributions**
    - *(同上)*
- **3.3 Zhang Olivia's Contributions**
    - *(同上)*
- **3.4 Wong Haw's Contributions**
    - *(同上)*

Part 4: References (参考文献)

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- *(在此处列出所有对项目有重要影响的外部文献、资料和网站链接。)*